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P5041

SEAT No. :
[Total No. of Pages : 2

T.E./Insem. - 539
T.E. (Computer Engg.)
DATABASE MANAGEMENT SYSTEMS APPLICATIONS
(2012 Pattern) (Semester - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Answer Q1 Or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Answer suitable data, if necessary.*

Q1) a) What is index created on table column? How it improves performance of SELECT query? **[5]**

b) What is the impact of insert, update & delete anomaly on overall design of database? How normalization is used to remove these anomalies? **[5]**

OR

Q2) a) Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. **[5]**

b) Consider following schema : **[5]**

account(acct_no, branch_name, balance)

Depositor(cust_name, acct_no)

borrower(cust_name, loan_no)

Write following queries using SQL (any 2)

- i) Find all customers who are having an account and loan or both.
- ii) Find all customers who have an account but no loan at the bank.
- iii) Find average account balance at each branch.

P.T.O.

Q3) a) Consider following structure for MongoDB collection and write a query for following requirements in MongoDB (any 2) [5]

Emp (EmpNo, EName, Department, Designation, Salary)

Department(Dno, Dname)

- i) Create above collections and insert minimum 2 documents into collection.
 - ii) Display all the employees working in COMP department.
 - iii) Update the dname to IT for Dno 101. If such department is not available in collection make new entry.
- b) Explain aggregation in MongoDB with suitable Example. [5]

OR

Q4) a) Explain how NOSQL databases are different than relational databases? [5]

b) List NOSQL data Models and explain any one in detail. [5]

Q5) a) Define schedule. Explain with example recoverable schedule. [5]

b) What is the requirement of atomicity and durability property of relational databases? Explain with example. [5]

OR

Q6) a) Write note on Performance Tuning in NOSQL databases. [5]

b) What is transaction? Explain different states of transaction. [5]
